

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1674

Laurel City 138 kV
March 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1674. The Interconnection Customer proposes a 128.8 MW battery energy storage facility interconnecting at the Laurel City 138 kV substation in the North Carolina area. The request was received on February 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 579 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,491,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Spring Street / Chipola 138 KV	Mitchell Dam / Skaggs Island Jct 2 345 KV line	111.5	234	7.2
Syl Laskin / Mitchell Dam 138 KV line	Mitchell Dam / Haggin 345 KV L	109.8	228	6.8
Fluvanna P.S / Bear 345 KV line	Leavitt / Birchwood 345 KV L	108.3	1,663	15.9
Westcoast Cellufibre Substation / Leavitt 138 KV Ckt 1	Mitchell Dam / Skaggs Island Jct 2 345 KV	106.1	384	7.8
Westcoast Cellufibre / Leavitt 138 KV	Mitchell Dam / Skaggs Island Junction 2 345 KV line	106.1	384	7.8
Hardy Reit / Stepdown 138 KV line	Mira Vista / Yarnen 345 KV	104.6	408	5.6
Chancellor Row / Natchez 138 KV line	Skaggs Island Jct 2 / Mitchell Dam 345 KV L	102.7	508	6.3
Fritch / Big Hanaford 138 KV	Skaggs Island Jct 2 / Mitchell Dam 345 KV	102.5	823	8.8
Lakeside Substation / Natchez 138 KV line	Mitchell Dam / Skaggs Island Jct 2 345 KV	102.3	531	6.3
Fritch / Lakeside 138 KV	Mitchell Dam / Skaggs Island Junction 2 345 KV	101.2	607	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chipola / Spring St 138 KV	Rebuild line with higher-capacity conductor	\$11,032,000	18% (\$2,039,000)
Syl Laskin / Mitchell Dam 138 KV line	Reconductor line	\$935,000	19% (\$177,000)
Bear Substation / Fluvanna P.S 345 KV Line	Reconductor line	\$31,077,000	5% (\$1,534,000)
Westcoast Cellufibre Substation / Leavitt 138 KV Ckt 1	Reconductor line	\$579,000	18% (\$107,000)
Leavitt / Westcoast Cellufibre 138 KV #2	Reconductor line	\$1,229,000	18% (\$228,000)
Hardy Reit / Stepdown 138 KV line	Reconductor line	\$2,920,000	22% (\$644,000)
Chancellor Row / Natchez 138 KV line	Reconductor line	\$1,129,000	18% (\$208,000)
Fritch / Big Hanaford 138 KV	Reconductor line	\$1,504,000	16% (\$235,000)
Lakeside / Natchez Substation 138 KV	Reconductor line	\$619,000	18% (\$114,000)
Fritch / Lakeside 138 KV	Reconductor line	\$1,112,000	18% (\$205,000)
	Total allocated to project		\$5,491,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.